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CalibrationDefTopic

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About Calibration

About Calibration

Calibration is the process of measuring the performance of the test system, as a whole or by instrument, and making whatever adjustments are needed to ensure that the system performs according to specifications. Changes in environmental conditions and configuration can necessitate recalibration. Since component performance may drift over time, calibration values are not permanent and must be re-established periodically.

To meet ISO requirements, you must ensure that your test system performs according to specification. An uncalibrated system does not meet ISO requirements because it does not guarantee that parts are tested to a known set of standards. IG-XL enforces proper calibration schedules where it is possible to do so.

You can run calibration as often as you like, but the process takes time. To minimize disruption of test floor throughput, IG-XL performs calibration only when necessary. The software automatically schedules and performs calibration of internal test system components; this is called [automatic calibration](#), or autocal. Autocal efficiently balances the need to keep the test system operating according to specification with the need to minimize interruption of testing.

[Some aspects of the test system, however, require user intervention, or manual calibration.](#) Many UltraFLEX instruments require manual calibration only infrequently, although the intervals vary according to instrument.

[You can control some aspects of calibration, while IG-XL manages others with little or no interaction.](#) Knowledge of how calibration works on your test system, however, can help you manage your test programs and device testing efficiently and make informed choices when the software asks you for input.

[This section covers the following information:](#)

- | |
|---|
| <ul style="list-style-type: none">• Calibration Terms and Definitions |
|---|

•	Levels of Calibration
•	How IG-XL Manages Calibration
•	How IG-XL Handles Calibration for a Job
•	Troubleshooting Calibration Problems
•	Calibration Types

Calibration information includes the following sections:

•	Automatic Calibration
•	Manual Calibration
•	Digital Instrument Calibration

For information on the calibration tools in the Maintenance Environment, see the [Help in the Maintenance Environment](#). To launch the Maintenance Environment on a standard IG-XL installation, select Start>Programs>Teradyne IG-XL Vx.yy.zz> Tools>Maintenance, where x.yy.zz is the version of IG-XL you are currently running. You must exit IG-XL before you can launch the Maintenance Environment.

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Automatic Calibration

Automatic Calibration

Automatic calibration, or autocal, schedules and performs calibration of internal test system components. Autocal efficiently balances the need to keep the test system operating according to specification with the need to minimize interruption of testing.

Differences Between Automatic and Manual Calibration

Automatic calibration differs from manual calibration in the following ways:

- Autocal requires no user intervention; it uses internal reference standards, which are not available to user software. Manual calibration uses external standards and requires that you perform some action, such as installing a calibration DIB.
- Autocal does not affect the DIB or the device and is, therefore, insufficient for calibrating attributes of a DIB, such as user DIB calibration. Manual calibration can extend to the device socket.

For more information on calibration, see:

- How Automatic Calibration Works
- Controlling Autocal from a Test Program
- When Autocal Runs
- Autocal Run Delays

•	About Calibration
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•	Manual Calibration
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Automatic Calibration : How Automatic Calibration Works

How Automatic Calibration Works

IG-XL checks the calibration status of each channel in your channel map before you run a test program for the first time and again before each subsequent run of the same program. (Note that calibration is relative to each channel, not the entire board.) For a discussion of changes that trigger calibration, see [When Autocal Runs](#).

If a channel is out of calibration, IG-XL takes the following actions:

- | | |
|----|--|
| 1. | If the channel requires calibration because the temperature has changed, and the calibration database includes calibration factors for the new temperature, IG-XL loads the stored calibration factors. |
| 2. | If the maximum allowable time between calibrations has passed, or if the temperature has changed but the database does not include calibration factors for the new temperature, IG-XL calls the appropriate calibration agent to perform calibration, then stores the new factors in the database. |

This requires no interaction with the test system operator.

Autocal works with test system boards and instruments. It does not affect the DIB or your device. Therefore it does not require undocking during production.

If a board or instrument fails calibration, IG-XL alerts you and prevents further runs of the test program until all test system elements pass. In engineering mode, IG-XL lets you use nominal (default) calibration factor values to permit program development. Take care, however, when interpreting results generated using nominal calibration factors.

For more information, see:

- [How IG-XL Manages Calibration](#)
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- [How IG-XL Handles Calibration for a Job](#)

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Automatic Calibration : Controlling Autocal from a Test Program

Controlling Autocal from a Test Program

You can use the following interpose functions to control the behavior of your test program before and after autocal runs:

- OnAutoCalStarted runs before IG-XL begins an autocal
- OnAutoCalCompleted runs after autocal finishes

For example, you can use these functions to display a notice in the production environment so engineers are aware that calibration is running and the tester computer has not locked up. If autocal takes no action, it does not invoke either of these functions. For more information, see:

- Interpose Functions Parameters in the VBT Test Library section
- Exec Interpose Functions in DataTool section

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Automatic Calibration : When Autocal Runs

When Autocal Runs

The following factors can cause a channel to require calibration:

- [Test system temperature has changed significantly \(see Calibration Types for a list of allowable temperature shifts\)](#)
- [Instrument drift exceeds specified tolerances](#)
- [The maximum allowable time between calibrations has passed](#)

UltraFLEX monitors test system temperature using a sensor located in an air inlet on the support boards.

Some other factors can require that you perform calibration. These include replacing the DIB, adding or moving instruments in the test head, and time. These factors trigger manual calibration, which requires intervention by the test system operator.

For more information, see:

- [How Automatic Calibration Works](#)
- [Differences Between Automatic and Manual Calibration](#)
- [About Calibration](#)
- [Manual Calibration](#)

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Automatic Calibration : Autocal Run Delays

Autocal Run Delays

IG-XL checks the calibration status of each channel in your channel map between test program runs. If any channels are out of calibration, IG-XL performs automatic calibration (autocal) on those channels. (For more information, see [How Automatic Calibration Works](#).)

Because calibration takes some amount of time, you may need to account for this time in a production environment. For example, you may need to set handler time-out delays to allow sufficient time for autocal to complete.

Autocal runs serially on each required channel. To calculate the maximum autocal time required for a test program, add the autocal times for all channels in the test program.

The following table lists instrument calibration times. You can use this information to determine how long the channels in a test program may take to run autocal.

Instrument Calibration Times

Autocal Type	Calibration Time (seconds)	Unit
AWG	4	Instrument
AWG PPMU	3	Channel
BBAC Source and Capture	72	Channel
BBAC PPMU	24	Channel
DC30 DCDiffMeter	2	Channel
DC30 DCTime Interpolation	2	Channel
DC30 DCTime TrigLevel	61	Channel

DC30 DCVI	24	Channel
DC30 DCVI Merged Pair	5	Channel pair
DC75 DCTime Interpolation	2.8	Channel
DC75 DCTime TrigLevel	69	Channel
DC75 DCVI Merged Pair	15	Channel pair
DC75 DiffMeter	2.5	Channel
DC75 VI	35.7	Channel
DCIO	454	Instrument
GigaDig	10	Instrument
GigaDig PPMU	3	Channel
HDVS	10	Channel
HexVS Merged	97.4	Channel pair
HexVS Precision ADC and Current Source	10.5	Channel
HexVS VS	18	Channel
Support Board	4.4	Instrument
UltraWave12G Synth	150	Instrument
UltraWave12G Measure	120	Instrument
UltraWave24G Synth	220	Instrument
UVS256	6	Channel
UVS256 Merged2	2	2 Merged Channels
UVS256 Merged4	1.5	4 Merged Channels
UVS256 Merged8	1.5	8 Merged Channels
VHFAC Capture DDS Timings	13	Channel
VHFAC Capture DDS Timings	13	Channel
VHFAC PPMU	24	Channel
VHFAC Serial Bus	8	Instrument
VHFAC Source DDS Timings	13	Channel
VHFAC Time	94	Channel
VSM	490	Instrument

Notes:

- All times are accurate to within one second.
- HPM, HSD1000, SB6G, UltraPin1600, UltraPin800, UltraSerial10G, UltraVI80, and UltraPAC80 do not perform automatic calibration.
- VHFAC Source performs leveling calibration during the first test program run. This calibration takes approximately 100 ms.



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About Calibration : Calibration Terms and Definitions

Calibration Terms and Definitions

IG-XL calibration uses the following terms:

autocal	Automatic calibration, scheduled and performed by IG-XL. Autocal works with boards and test system components only and requires no user intervention. Autocal does not affect the DIB or the DUT and, therefore, does not require undocking.
calibration agent	Driver-specific software that handles calibration tasks for a type of test system board.
calibration correction	Value by which an instrument adjusts your programming of a signal to make sure that the signal arrives at the device when you expect it to arrive. This may be the same as a calibration factor, or the factors may be manipulated to compute the adjustment. This is of interest only if you manually adjust your results or the values you program; otherwise, IG-XL manages this information.
calibration database	Central repository for all calibration packets for a specific test system. The Calibration Service maintains this database. For more information, see Calibration Database.
calibration DIB	Teradyne-supplied DIB used only for performing calibration tasks.
calibration factor	Value stored in a calibration packet for the purpose of adjusting the instrument. This typically adjusts voltage, current, or time.
calibration measurement	Measurement made while calibration is running. This can become a calibration factor, or multiple measurements may be averaged or

	summed or otherwise manipulated to compute the stored factor. This is of interest only if you create your own calibration routine; otherwise, IG-XL manages this information.
calibration packet	Set of data containing the conditions and calibration factor for one attribute of a test system component. Calibration packets consist of a header and data. For more information, see How IG-XL Manages Calibration.
Calibration Service	Part of IG-XL that runs in the background to oversee test system calibration. When necessary, it performs autocal or alerts you to the need for manual calibration.
external calibration	Calibration that requires user intervention.
internal calibration	Calibration that does not require user intervention.

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About Calibration : Levels of Calibration

Levels of Calibration

The levels of calibration are:

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|----|--|
| 1. | Internal (autocalibration, or autocal). IG-XL calibrates instruments and channels against internal standards. This does not require that you take any action; IG-XL performs all management. To calibrate specific instruments or test system resources, use the Maintenance Environment. For more information, see External Calibration. |
| 2. | Self-test and performance verification (checkers). |
| 3. | External calibration (calibration DIB or user DIB). IG-XL calibrates internal standards against fixed, external standards. IG-XL can notify you when this is due, but you must interact with the tester or the software to perform external calibration. For more information on using a calibration DIB or calibrating the DIB for your device, see External Calibration. |

You must complete levels 1 through 3 before any test program run.

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About Calibration : How IG-XL Manages Calibration

How IG-XL Manages Calibration

IG-XL calibration has two nonhardware components: the calibration database, and the calibration software.

Calibration Database

IG-XL stores calibration information in a calibration database. The smallest usable piece of information is called a calibration packet, or “cal packet.” A cal packet has two pieces: the header and the data.

The header is stored in a database file and contains searchable identification information. IG-XL uses this header to locate cal packets according to the serial number of a particular board, a test system temperature reading, the most recent recalibration date, or other criteria.

The data is stored in a separate file and contains calibration information. This includes calibration factors or other information specific to the type of board or type of calibration.

IG-XL stores these files in two locations under the IG-XL folder. (If you performed a standard installation, this is C:\Program Files (x86)\IG-XL.) The subfolders are:

Cal	Contains the calibration factor files, one for each board type.
MSSQL\data	Contains the calibration database that stores the headers for each cal packet.

Note:

Do not delete or edit any files in these folders. Missing or corrupt calibration data can cause calibration to fail or can lead to test system instability.

Cal packets are valid for a specified length of time. If that time passes without recalibration, the cal packet “expires” and forces a recalibration. The length of time a cal packet is valid varies according to instrument.

When a cal packet expires, it remains in the database but is marked invalid. This allows for trend analysis using historical data.

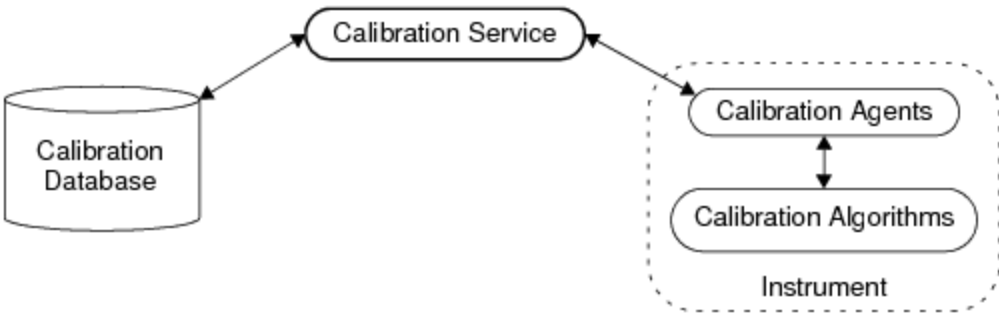
The Calibration Service maintains the calibration database.

Calibration Data Reuse

To minimize the time spent on test system calibration, IG-XL reuses calibration factors for as long as the factors are valid. When you first calibrate the test system for a specific set of conditions (such as temperature and configuration), IG-XL stores the resulting calibration factors in a Calibration Database. The next time IG-XL encounters identical conditions, it does not recalibrate the entire test system; instead, it uses the calibration factors stored in the database. These values load in much less time than it takes to perform another calibration, which improves overall tester utilization.

Calibration Software

The figure describes the relationships among the pieces of calibration software.



Distribution of Calibration Responsibilities

The Calibration Service monitors the status for all relevant cal packets. When a cal packet becomes invalid for any reason, the Service invokes a calibration agent to perform the needed calibration and update the database. These agents use calibration algorithms to determine cal factors for each aspect of the board.

The calibration packet header includes a revision number. The number changes only when the format changes, not at each software revision. This facilitates loading the correct calibration packets for the version of IG-XL you are running. IG-XL ignores cal packets from versions it does not recognize (that is, later versions). Keeping data from different revisions in the same database allows you to switch IG-XL versions without having to recalibrate the entire test system each time.

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About Calibration : How IG-XL Handles Calibration for a Job

How IG-XL Handles Calibration for a Job

Before you run or loop a job, you must perform calibration to ensure that the test system operates according to specification.

Note:

After you cycle power, let your test system warm up for 30 minutes. This ensures that all test system temperatures are within specifications. If you do not observe this warm-up time, the tester will not meet specified accuracy.

IG-XL takes these steps to verify correct calibration:

1. If you are using HSD channels with a DIB, you must calibrate the DIB for all HSD channels in your channel map. Do this after you load your test program.

This configures each digital channel with the correct path length information and propagation delays for the traces on your DIB.

Note that this type of calibration requires user interaction. For information on performing user DIB calibration, see [User DIB Calibration](#).

2. When you load and validate your test program, IG-XL reviews the channel map to determine what test system resources require calibration. You must have valid calibration factors for all channels before you run your test program.

IG-XL compiles a list of channels but performs no calibration at validation time.

Note that IG-XL calibrates according to your channel map only; to perform a full calibration before a test program run, use the Calibration tab in the Maintenance Environment. For more information, see [Maintenance Environment’s Calibration Tab](#).

3. Before you run your test program for the first time, IG-XL checks the calibration status for all channels in the channel map. It uses the list created at validation. For each channel in the list:

- If the [calibration database](#) contains valid calibration factors, IG-XL uses the factors from the database.
- If the database contains no valid calibration factors, the [Calibration Service](#) calibrates that channel and stores the resulting cal factors for future reference.

In debug (engineering) mode you can tell IG-XL how to proceed if calibration fails or for any reason cannot provide all the required calibration factors. The options are:

- Use nominal (default) values; this permits test program runs and is, therefore, useful for debugging test program flow, but it does not produce reliable results
- Abort the job

☑ Note:

In production mode, a job cannot run if any calibration failures exist or if any calibration data is missing or invalid.

4. Before each subsequent run, the Calibration Service monitors test system temperature, the expiration time for calibration packets relevant to channels your test program uses, and other additional information as needed by the test system instrumentation.

If any calibration packets have expired, the Calibration Service invokes calibration for these channels (packets).

If the test system temperature has changed by more than the allowable margin, the Calibration Service tries to load calibration factors for the new temperature from the database. The Service invokes autocal for any attributes that do not have valid factors.

5. If you add to or change your channel map, the Calibration Service launches autocal for any additional or modified channels. If you remove channels from your channel map, no additional calibration is required.

For instrument-specific calibration considerations, click [Related Topics](#).

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About Calibration : Troubleshooting Calibration Problems

Troubleshooting Calibration Problems

If calibration fails, you can check two log files for possible causes of the failure.

The first place to look when investigating calibration failures is the calibration log file.

IG-XL logs calibration events over time. It keeps track of such events as when a board was calibrated and what results that calibration run produced. The log file contains most of the messages that appear in the Calibration Output Window and also various responses and messages from IG-XL and from instrument drivers. This can help you isolate the source of a calibration problem.

IG-XL stores the calibration log file as calibration.log in the \tester folder for each version of IG-XL you have installed.

You can also look in the log of run-time errors. This file is not specific to calibration but does capture any failure messages that calibration generates. The information in this file does not appear in the Output Window.

IG-XL stores the run-time error log as RuntimeErrors.log in the \tester folder for each version of IG-XL you have installed.

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About Calibration : Calibration Types

Calibration Types

The table below describes the types of calibration available for each instrument, where:

Board Name	Instrument to which the specific calibration applies.
Calibration Type	Scope of calibration (internal or external); for external calibration, includes the type of DIB required.
Calibration ID	Name for this type of calibration in the Maintenance Environment.
Temp Delta ()	Temperature variance over which a specific calibration is valid. A change in temperature greater than this number requires recalibration. Values are in degrees Celsius.
Validity (days)	Length of time this calibration is valid. Passage of this amount of time since the last calibration requires recalibration, even if no other conditions (for example, temperature) change.

When one of these items exceeds the limits listed in the following table, IG-XL performs automatic calibration for this item. This calibration occurs between test program runs, so it does not interrupt test program flow.

All calibration factors are stored on disk on the test system computer except DCCalExternal for the SupportBoards and any factory calibration factors, which are stored on the boards themselves.

Calibration Types for Test System Components

Board Name	Calibration Type	Calibration ID	Temp Delta (Degree C)	Validity (days)
AWG6G	Internal	AWGPBa_1	1.7	90
AWG6G	Internal	AWGPBa_ADC_1	3	90
AWG6G	Internal	AWGPBa_PPMU_1_NEG	3	90
AWG6G	Internal	AWGPBa_PPMU_1_POS	3	90
AWG6G	Internal	AWGPBa_PPMU_2_NEG	3	90
AWG6G	Internal	AWGPBa_PPMU_2_POS	3	90
AWG6G	Internal	AWGPBa_PPMU_3_NEG	3	90
AWG6G	Internal	AWGPBa_PPMU_3_POS	3	90
AWG6G	Internal	AWGPBa_PPMU_4_NEG	3	90
AWG6G	Internal	AWGPBa_PPMU_4_POS	3	90
BBAC-15	Internal	BBAC_15_Cap_1	3	7
BBAC-15	Internal	BBAC_15_Cap_1	3	7
BBAC-15	Internal	BBAC_15_Cap_2	3	7
BBAC-15	Internal	BBAC_15_Cap_2	3	7
BBAC-15	Internal	BBAC_15_CapNeg_PPMU_1	3	7
BBAC-15	Internal	BBAC_15_CapNeg_PPMU_1	3	7
BBAC-15	Internal	BBAC_15_CapNeg_PPMU_2	3	7
BBAC-15	Internal	BBAC_15_CapNeg_PPMU_2	3	7
BBAC-15	Internal	BBAC_15_CapPos_PPMU_1	3	7
BBAC-15	Internal	BBAC_15_CapPos_PPMU_1	3	7
BBAC-15	Internal	BBAC_15_CapPos_PPMU_2	3	7
BBAC-15	Internal	BBAC_15_CapPos_PPMU_2	3	7
BBAC-15	Internal	BBAC_15_Src_1	3	7
BBAC-15	Internal	BBAC_15_Src_1	3	7
BBAC-15	Internal	BBAC_15_Src_2	3	7
BBAC-15	Internal	BBAC_15_Src_2	3	7

BBAC-15	Internal	BBAC_15_SrcNeg_PPMU_1	3	7
BBAC-15	Internal	BBAC_15_SrcNeg_PPMU_1	3	7
BBAC-15	Internal	BBAC_15_SrcNeg_PPMU_2	3	7
BBAC-15	Internal	BBAC_15_SrcNeg_PPMU_2	3	7
BBAC-15	Internal	BBAC_15_SrcPos_PPMU_1	3	7
BBAC-15	Internal	BBAC_15_SrcPos_PPMU_1	3	7
BBAC-15	Internal	BBAC_15_SrcPos_PPMU_2	3	7
BBAC-15	Internal	BBAC_15_SrcPos_PPMU_2	3	7
BBAC-15	Internal	BBAC_15_SrcRef_PPMU_1	3	7
BBAC-15	Internal	BBAC_15_SrcRef_PPMU_1	3	7
BBAC-15	Internal	BBAC_15_SrcRef_PPMU_2	3	7
BBAC-15	Internal	BBAC_15_SrcRef_PPMU_2	3	7
BBAC-15	Internal	BBAC_15_SrcRef_PPMU_2	3	7
DC-30	Internal	DC30Cal Internal VI1_2 Merged	5	180
DC-30	Internal	DC30Cal Internal VI1_2 Merged	5	180
DC-30	Internal	DC30Cal Internal VI11_12 Merged	5	180
DC-30	Internal	DC30Cal Internal VI11_12 Merged	5	180
DC-30	Internal	DC30Cal Internal VI13_14 Merged	5	180
DC-30	Internal	DC30Cal Internal VI13_14 Merged	5	180
DC-30	Internal	DC30Cal Internal VI15_16 Merged	5	180
DC-30	Internal	DC30Cal Internal VI15_16 Merged	5	180
DC-30	Internal	DC30Cal Internal VI17_18 Merged	5	180
DC-30	Internal	DC30Cal Internal VI17_18 Merged	5	180

DC-30	Internal	DC30Cal Internal VI19_20 Merged	5	180
DC-30	Internal	DC30Cal Internal VI19_20 Merged	5	180
DC-30	Internal	DC30Cal Internal VI3_4 Merged	5	180
DC-30	Internal	DC30Cal Internal VI3_4 Merged	5	180
DC-30	Internal	DC30Cal Internal VI5_6 Merged	5	180
DC-30	Internal	DC30Cal Internal VI5_6 Merged	5	180
DC-30	Internal	DC30Cal Internal VI7_8 Merged	5	180
DC-30	Internal	DC30Cal Internal VI7_8 Merged	5	180
DC-30	Internal	DC30Cal Internal VI9_10 Merged	5	180
DC-30	Internal	DC30CalInternalDVM1	5	180
DC-30	Internal	DC30CalInternalDVM2	5	180
DC-30	Internal	DC30CalInternalTime1Intrpl	10	90
DC-30	Internal	DC30CalInternalTime1TrigLvl	5	180
DC-30	Internal	DC30CalInternalTime2Intrpl	10	90
DC-30	Internal	DC30CalInternalTime2TrigLvl	5	180
DC-30	Internal	DC30CalInternalVI1	5	180
DC-30	Internal	DC30CalInternalVI10	5	180
DC-30	Internal	DC30CalInternalVI11	5	180
DC-30	Internal	DC30CalInternalVI12	5	180
DC-30	Internal	DC30CalInternalVI13	5	180
DC-30	Internal	DC30CalInternalVI14	5	180
DC-30	Internal	DC30CalInternalVI15	5	180
DC-30	Internal	DC30CalInternalVI16	5	180
DC-30	Internal	DC30CalInternalVI17	5	180
DC-30	Internal	DC30CalInternalVI18	5	180
DC-30	Internal	DC30CalInternalVI19	5	180
DC-30	Internal	DC30CalInternalVI2	5	180

DC-30	Internal	DC30CalInternalVI20	5	180
DC-30	Internal	DC30CalInternalVI3	5	180
DC-30	Internal	DC30CalInternalVI4	5	180
DC-30	Internal	DC30CalInternalVI5	5	180
DC-30	Internal	DC30CalInternalVI6	5	180
DC-30	Internal	DC30CalInternalVI7	5	180
DC-30	Internal	DC30CalInternalVI8	5	180
DC-30	Internal	DC30CalInternalVI9	5	180
DC-75	Internal	DC75Cal Internal VI1_2 Merged	5	180
DC-75	Internal	DC75Cal Internal VI3_4 Merged	5	180
DC-75	Internal	DC75CalInternalDVM1	5	180
DC-75	Internal	DC75CalInternalDVM2	5	180
DC-75	Internal	DC75CalInternalTime1Intrpl	10	90
DC-75	Internal	DC75CalInternalTime1TrigLvl	5	180
DC-75	Internal	DC75CalInternalTime2Intrpl	10	90
DC-75	Internal	DC75CalInternalTime2TrigLvl	5	180
DC-75	Internal	DC75CalInternalVI1	5	180
DC-75	Internal	DC75CalInternalVI2	5	180
DC-75	Internal	DC75CalInternalVI3	5	180
DC-75	Internal	DC75CalInternalVI4	5	180
DCIOBoard	External	Independent	None	100
GigaDigHD2	Internal	GigaDigPBa_1	3	3
GigaDigHD2	Internal	GigaDigPBa_2	3	3
GigaDigHD2	Internal	GigaDigPBa_3	3	3
GigaDigHD2	Internal	GigaDigPBa_4	3	3
GigaDigHD2	Internal	GigaDigPBa_5	3	3
GigaDigHD2	Internal	GigaDigPBa_6	3	3
GigaDigHD2	Internal	GigaDigPBa_7	3	3
GigaDigHD2	Internal	GigaDigPBa_8	3	3

GigaDigHD2	Internal	GigaDigPBa_PPMU_1_NEG	3	3
GigaDigHD2	Internal	GigaDigPBa_PPMU_1_POS	3	3
GigaDigHD2	Internal	GigaDigPBa_PPMU_2_NEG	3	3
GigaDigHD2	Internal	GigaDigPBa_PPMU_2_POS	3	3
GigaDigHD2	Internal	GigaDigPBa_PPMU_3_NEG	3	3
GigaDigHD2	Internal	GigaDigPBa_PPMU_3_POS	3	3
GigaDigHD2	Internal	GigaDigPBa_PPMU_4_NEG	3	3
GigaDigHD2	Internal	GigaDigPBa_PPMU_4_POS	3	3
GigaDigHD2	Internal	GigaDigPBa_PPMU_5_NEG	3	3
GigaDigHD2	Internal	GigaDigPBa_PPMU_5_POS	3	3
GigaDigHD2	Internal	GigaDigPBa_PPMU_6_NEG	3	3
GigaDigHD2	Internal	GigaDigPBa_PPMU_6_POS	3	3
GigaDigHD2	Internal	GigaDigPBa_PPMU_7_NEG	3	3
GigaDigHD2	Internal	GigaDigPBa_PPMU_7_POS	3	3
GigaDigHD2	Internal	GigaDigPBa_PPMU_8_NEG	3	3
GigaDigHD2	Internal	GigaDigPBa_PPMU_8_POS	3	3
HexVS	Internal	HexVSCalMIMerge	3	30
HexVS	Internal	HexVSCalPadclsrc	3	30
HexVS	Internal	HexVSCalVS1	3	30
HexVS	Internal	HexVSCalVS2	3	30
HexVS	Internal	HexVSCalVS3	3	30
HexVS	Internal	HexVSCalVS4	3	30
HexVS	Internal	HexVSCalVS5	3	30
HexVS	Internal	HexVSCalVS6	3	30
HSD-I	External	Independent	None	93
HSD-I	External (User DIB)	JobDependent	None	Infinite ¹
HSD-M	External	Independent	None	93

HSD-M	External (User DIB)	JobDep	None	Infinite ¹
HSD-U	External	Independent	None	93
HSD-U	External (User DIB)	JobDep	None	Infinite ¹
HSS-6400	External	JobIndependent	None	93
MWMeasure	External	BurnNVRAM	3	Infinite ¹
MWMeasure	Internal	LevelingFlush	3	1/3
MWMeasure	External	PowerTransfer	3	365
MWMeasure	External	ReceiverEqualization	3	90
MWMeasure	External	ReceiverGain	3	90
MWMeasure	External	SourceAddPath	0	90
MWMeasure	External	SourceAttenuator	3	90
MWMeasure	External	SourceSetup	3	90
MWMeasure	External	SourceSetupAuto	3	7
MWMeasure	External	SourceVCA	0	90
MWMeasure	External	ThermalTrim	3	365
MWMeasure	External	VNA	3	365
MWMeasureHD	External	SourceVCA	0	90
MWMeasureHD	External	SourceGainTrim	0	Infinite ¹
MWMeasureHD	External	SourceAttenuator	0	90
MWMeasureHD	External	ReceiverGain	7	90
MWMeasureHD	External	ReceiverEqualization	0	90
MWMeasureHD	External	VNA	0	360
MWMeasureHD	External	PowerTransfer	0	360
MWMeasureHD	External	BurnNVRAM	0	Infinite ¹
MWSynthHD	Internal	YIGCoarseTune	7	90
MWSynthHD	Internal	SynthLOLevel	7	90
MWSynthHD	Internal	SynthSignalLevel	7	90

MWSynthHD	Internal	SynthSideband	0	Infinite ¹
MWSynthHD	Internal	SynthSpurious	1	3
MWSynthHD	Internal	DivModInputLevel	3	7
MWSynthHD	Internal	ModulatorStaticBalance	3	7
MWSynthHD	Internal	DivModLevelTrim	3	90
MWSynthHD	External	DUTClockCal	0	Infinite ¹
MWSynthHD	External	BurnNVRAM	0	Infinite ¹
MWSynthRef	External	BurnNVRAM	3	Infinite ¹
MWSynthRef	External	DUTClockLevel	0	Infinite ¹
MWSynthRef	Internal	DutClockLevelTrim	3	7
MWSynthRef	External	DUTLOLevel	0	Infinite ¹
MWSynthRef	Internal	DutLoLevelTrim	3	7
MWSynthRef	External	ModInputLevel	0	Infinite ¹
MWSynthRef	External	ModOutputLevel	0	Infinite ¹
MWSynthRef	Internal	ModOutputLevelTrim	25	Infinite ¹
MWSynthRef	Internal	ModulatorStaticBalance	0	7
MWSynthRef	Internal	PLLOutputLevel	3	7
MWSynthRef	External	RcvLOLevel	0	Infinite ¹
MWSynthRef	Internal	SynthLOLevel	7	90
MWSynthRef	External	SynthSideband	0	Infinite ¹
MWSynthRef	Internal	SynthSignalLevel	7	90
MWSynthRef	Internal	SynthSpurious	1	3
MWSynthRef	Internal	YIGCoarseTune	7	7
SupportBoard	External	DCExt	None	186
SupportBoard	Internal	DCInt	1.5	1
Turbo-Base	Internal	BBAC_15_Cap_1	3	7
Turbo-Base	Internal	BBAC_15_Cap_2	3	7
Turbo-Base	Internal	BBAC_15_CapNeg_PPMU_1	3	7

Turbo-Base	Internal	BBAC_15_CapNeg_PPMU_2	3	7
Turbo-Base	Internal	BBAC_15_CapPos_PPMU_1	3	7
Turbo-Base	Internal	BBAC_15_CapPos_PPMU_2	3	7
Turbo-Base	Internal	BBAC_15_Src_1	3	7
Turbo-Base	Internal	BBAC_15_Src_2	3	7
Turbo-Base	Internal	BBAC_15_SrcNeg_PPMU_1	3	7
Turbo-Base	Internal	BBAC_15_SrcNeg_PPMU_2	3	7
Turbo-Base	Internal	BBAC_15_SrcPos_PPMU_1	3	7
Turbo-Base	Internal	BBAC_15_SrcPos_PPMU_2	3	7
Turbo-Base	Internal	BBAC_15_SrcRef_PPMU_1	3	7
Turbo-Base	Internal	BBAC_15_SrcRef_PPMU_2	3	7
Turbo-Frame	Internal	Turbo_Base_Cap_1	3	7
Turbo-Frame	Internal	Turbo_Base_Cap_2	3	7
Turbo-Frame	Internal	Turbo_Base_CapNeg_PPMU_1	3	7
Turbo-Frame	Internal	Turbo_Base_CapNeg_PPMU_2	3	7
Turbo-Frame	Internal	Turbo_Base_CapPos_PPMU_1	3	7
Turbo-Frame	Internal	Turbo_Base_CapPos_PPMU_2	3	7
Turbo-Frame	Internal	Turbo_Base_Src_1	3	7
Turbo-Frame	Internal	Turbo_Base_Src_2	3	7
Turbo-Frame	Internal	Turbo_Base_SrcNeg_PPMU_1	3	7
Turbo-Frame	Internal	Turbo_Base_SrcNeg_PPMU_2	3	7
Turbo-Frame	Internal	Turbo_Base_SrcPos_PPMU_1	3	7
Turbo-Frame	Internal	Turbo_Base_SrcPos_PPMU_2	3	7
Turbo-Frame	Internal	Turbo_Base_SrcRef_PPMU_1	3	7
Turbo-Frame	Internal	Turbo_Base_SrcRef_PPMU_2	3	7
Turbo-Frame	Internal	Turbo_Frame_SrcCap_1	3	7
Turbo-Frame	Internal	Turbo_Frame_SrcCap_2	3	7
Turbo-Frame	Internal	Turbo_SE_Linearity_1	3	7
Turbo-Frame	Internal	Turbo_SE_Linearity_2	3	7

VHDVS	Internal	VHDVS_CALID_Merge1	5	90
VHDVS	Internal	VHDVS_CALID_Merge2	5	90
VHDVS	Internal	VHDVS_CALID_Merge4	5	90
VHDVS	Internal	VHDVS_CALID_Merge8	5	90
VHFAC	Internal	VHFAC_CAD00	3	30
VHFAC	Internal	VHFAC_CAD01	3	30
VHFAC	Internal	VHFAC_CAP00	3	30
VHFAC	Internal	VHFAC_CAP01	3	30
VHFAC	Internal	VHFAC_PPMUC0PN	3	30
VHFAC	Internal	VHFAC_PPMUC0PP	3	30
VHFAC	Internal	VHFAC_PPMUC0SN	3	30
VHFAC	Internal	VHFAC_PPMUC0SP	3	30
VHFAC	Internal	VHFAC_PPMUC1PN	3	30
VHFAC	Internal	VHFAC_PPMUC1PP	3	30
VHFAC	Internal	VHFAC_PPMUC1SN	3	30
VHFAC	Internal	VHFAC_PPMUC1SP	3	30
VHFAC	Internal	VHFAC_PPMUS0PN	3	30
VHFAC	Internal	VHFAC_PPMUS0PP	3	30
VHFAC	Internal	VHFAC_PPMUS0SN	3	30
VHFAC	Internal	VHFAC_PPMUS0SP	3	30
VHFAC	Internal	VHFAC_PPMUS1PN	3	30
VHFAC	Internal	VHFAC_PPMUS1PP	3	30
VHFAC	Internal	VHFAC_PPMUS1SN	3	30
VHFAC	Internal	VHFAC_PPMUS1SP	3	30
VHFAC	Internal	VHFAC_PPMUSBCLK	3	30
VHFAC	Internal	VHFAC_PPMUSBIO0	3	30
VHFAC	Internal	VHFAC_PPMUSBIO1	3	30
VHFAC	Internal	VHFAC_PPMUTPN	3	30
VHFAC	Internal	VHFAC_PPMUTPP	3	30

VHFAC	Internal	VHFAC_PPMUTSN	3	30
VHFAC	Internal	VHFAC_PPMUTSP	3	30
VHFAC	Internal	VHFAC_SBC00	3	30
VHFAC	Internal	VHFAC_SRC00	3	30
VHFAC	Internal	VHFAC_SRC01	3	30
VHFAC	Internal	VHFAC_SRD00	3	30
VHFAC	Internal	VHFAC_SRD01	3	30
VHFAC	Internal	VHFAC_TIM00	3	30
VSM	Internal	DCCal	None	30

1 This calibration value does not trigger recalibration and therefore does not have an expiration time.

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Digital Instrument Calibration

Digital Instrument Calibration
Calibration for UltraFLEX digital instrument boards (HSD1000, UltraPin800, UltraPin1600, UltraPin4000, and HPM) includes the following parts:

- Independent Calibration
- Job Dependent Calibration

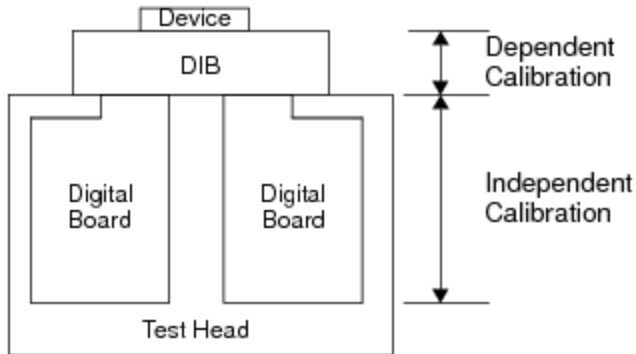
Dividing calibration maximizes efficiency by allowing you to calibrate only the parts of the hardware that require it. Each calibration covers a different part of the digital hardware and signal path. This section includes the following topics:

- Running Independent Calibration
- Running Job Dependent Calibration
- Calibration Dependencies
- Nominal Values
- Calibration Data Storage and Status Displays

Independent Calibration

Independent calibration calibrates test system hardware so that calibration data is not dependent on specific test program levels or timing and applies to all use conditions.

Independent calibration runs tests that are not DIB-dependent, but you cannot run it while a DIB is present because it conducts TDR measurements. You must remove the DIB from the test head to calibrate all levels and timing to the end of the JIB without failing. As shown in the figure below, independent calibration calibrates the digital board hardware only to the end of the JIB, while dependent calibration includes only traces on the DIB.



Independent and Dependent Calibration

Independent calibration is slot-dependent, so if a calibrated instrument is moved to another test head slot, you must recalibrate it. If you move the instrument back to the slot in which it was originally calibrated, however, then calibration is still valid.

Independent calibration is required every 93 days, or whenever a digital board is moved to a new location in the test head. If a digital board is moved to a slot where it has already been successfully calibrated within the past 93 days, you need not recalibrate the board.

Digital board job independent calibration takes the following measurements:

- Analog-to-Digital Converter (ADC)

- PPMU

- PE Drive and Compare Levels

- Timing Linearity

- Channel Deskew

When calibration is complete, the calibration values are stored in the host computer and PSP memory for each digital channel. The calibration values are reloaded into PSP after each power-up. For run-time calibration, the PSP loads calibration factors into ASIC registers before a pattern burst and may also make calculations for new levels.

For more information, see [Running Independent Calibration](#).

 Note:

The HSD1000 and UltraPin800 support an enhanced timing accuracy option that requires a special license and calibration DIB before running independent calibration on all HSD1000 or UltraPin800 boards in the test system. For more information, see [HSD1000 and UltraPin800 Enhanced Timing Accuracy Calibration Process](#).

Job Dependent Calibration

Job dependent calibration (also known as “DIB calibration”) measures trace lengths for job channels on a user DIB. This is performed online after the job is loaded and validated.

Digital board job dependent calibration:

- | | |
|---|---|
| • | Measures DIB path length |
| • | Performs calculations to optimize accuracy for job DIB conditions |
| • | Interpolates calibration data to cover all levels |
| • | Stores calculation data in PSP |

When the DIB has an EEPROM, run job dependent calibration once, and IG-XL writes the calibration data to the EEPROM by default. If calibration data exists in the DIB EEPROM circuit, then the data will be loaded during job validation and job dependent calibration is not required.

When the DIB does not have an EEPROM, run job dependent calibration each time a job is loaded or when the channel map is changed to include channels that have not been calibrated. In this case, IG-XL send calibration data to PSP memory.

For more information, see [Running Job Dependent Calibration](#).

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Digital Instrument Calibration : Running Independent Calibration

Running Independent Calibration

Use the Maintenance User Interface (MUI) to execute independent calibration. If you are using the enhanced timing accuracy option, you also must run independent calibration on all digital boards in the test system.

The following topics are available:

- Maintenance User Interface
- Digital Board Calibration Debug Display
- HSD1000 and UltraPin800 Enhanced Timing Accuracy Calibration Process
- UltraPin4000 Slot-to-Slot Calibration Process

Maintenance User Interface

The MUI program provides a high level view of the calibration status for all boards in the test system. From this interface, you can select and execute specific calibration types on the desired hardware.

To start the MUI:

- Select the Start>Programs>Teradyne IG-XL>Tools>Maintenance.
- In the MUI window, click the Calibration tab (in the panel below the tool bars).

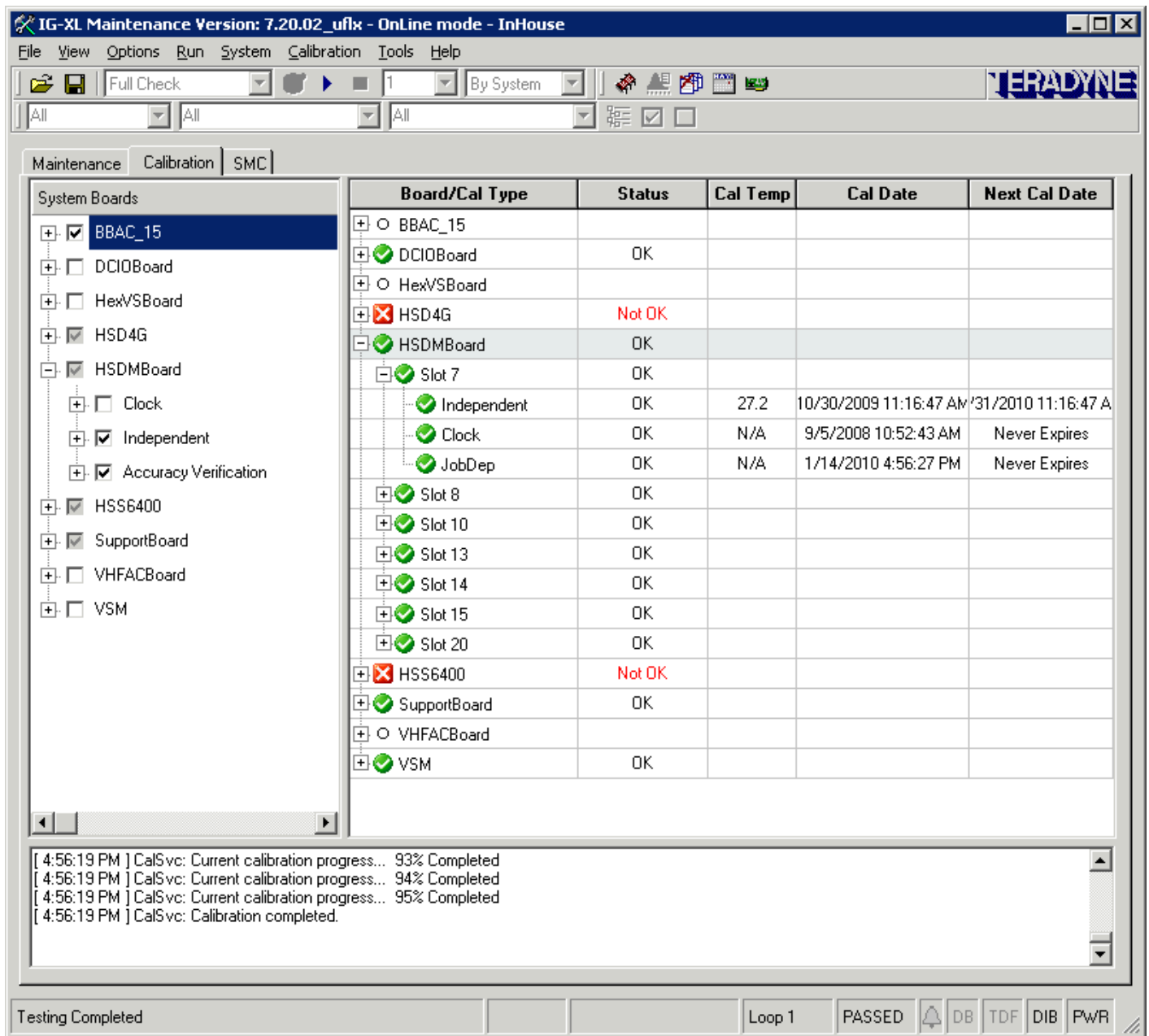
On the left side of the main calibration panel is a list of all hardware in the test system. The list is a tree that you can expand to show all the types of calibrations that can be run on that hardware. When Maintenance is started, calibrations that are currently required are automatically checked.

- | | |
|----|--|
| 3. | Check the boxes to select which calibrations to run and which boards to run them on. |
|----|--|

For example, you can choose to run only independent calibration on all HSD1000, UltraPin800, UltraPin1600, UltraPin4000, or HPM digital boards.

- | | |
|----|---|
| 4. | Click Run (located on the top toolbar). |
|----|---|

Calibration begins and a window appears showing all selected steps as they are executed. If any interaction is required during the calibration run, the window prompts you to perform the necessary actions.



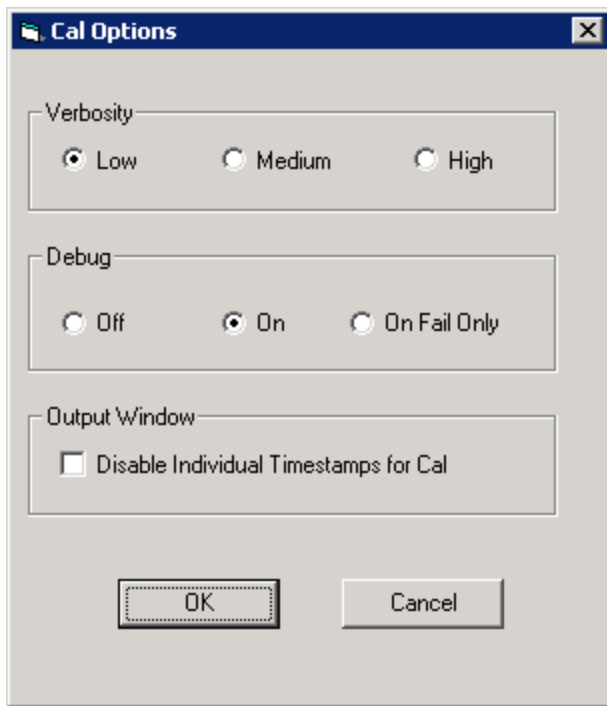
Maintenance User Interface Window

Digital Board Calibration Debug Display

To see a more detailed status of HSD1000, UltraPin800, UltraPin4000, UltraPin1600, or HPM calibration, you can optionally execute calibration from the digital board's Calibration Debug Display.

To open the debug display:

1. In the MUI, select Calibration>Options.
2. When the dialog box appears, turn the Debug option to On.



Activating the Calibration Debug Display

3. Click OK.

The Debug selection from the Calibration menu is now enabled.

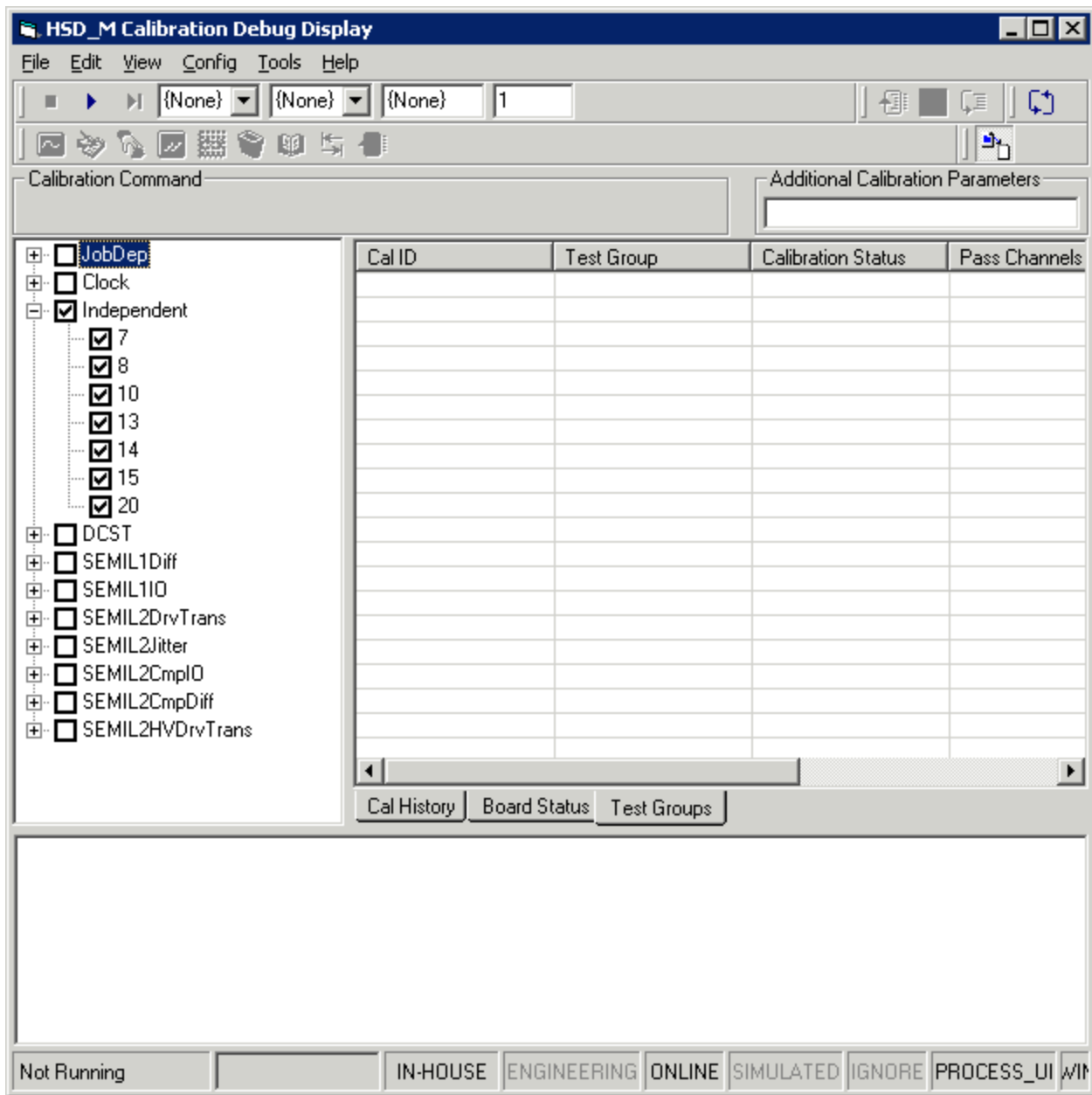
4. Select the Calibration menu again and select the digital board type (HSD1000, UltraPin800, UltraPin4000, UltraPin1600, or HPM) from the Debug submenu.

This opens the Calibration Debug Display.

5. Check the boxes to select which types of calibrations to run and which digital board slots to run them on.

The calibration messages window shows a detailed, customizable display of all messages generated during the calibration run.

Click the Test Groups tab from the calibration status panel (on the right hand side of the display) to see the pass/fail status of all channels for all calibration tests as they are executed.



Calibration Debug Display Window

HSD1000 and UltraPin800 Enhanced Timing Accuracy Calibration Process

The HSD1000 and UltraPin800 support an enhanced timing accuracy option that requires a special license and calibration DIB before running independent calibration on all HSD1000 or UltraPin800 boards in the test system. To enable this option follow these steps:

1. Install the license file provided by Teradyne into the following location:

C:\Program Files (x86)\Teradyne\IG-XL\Cal\vdig_license.dat (HSD1000)

C:\Program Files (x86)\Teradyne\IG-XL\Cal\idig_license.dat (UltraPin800)

You can order the enhanced accuracy license using Teradyne part number IN-406-00.

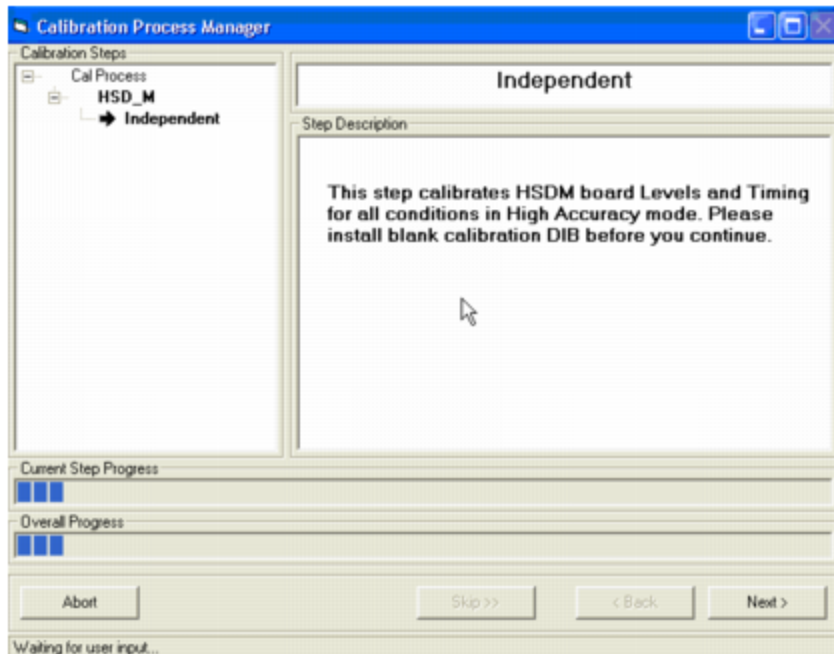
2. Install a special calibration DIB.

Teradyne part numbers for both the HSD1000 and UltraPin800 DIBs are 979-523-00 for 12 and 24-slot systems and 979-523-30 for 36-slot systems.

3. Once the license file and DIB are in place, run independent calibration in the Calibration tab of the IG-XL MUI.

If an incorrect DIB is installed or no DIB is present, then the Calibration Process Manager reminds you to install a valid calibration DIB, as shown in the figure below. Click Next to proceed.

If you click Next before installing a valid calibration DIB, then the DIB Down Detection calibration section fails, and the calibration run status reports EE (external equipment failure), rather than P (pass).



Calibration Process Manager Message for HSDM High Accuracy Mode

To disable this option, remove the vdig_license.dat (HSD1000) or idig_license.dat (UltraPin800) file from the IG-XL\Cal folder, restart the IG-XL MUI, and recalibrate the test system.

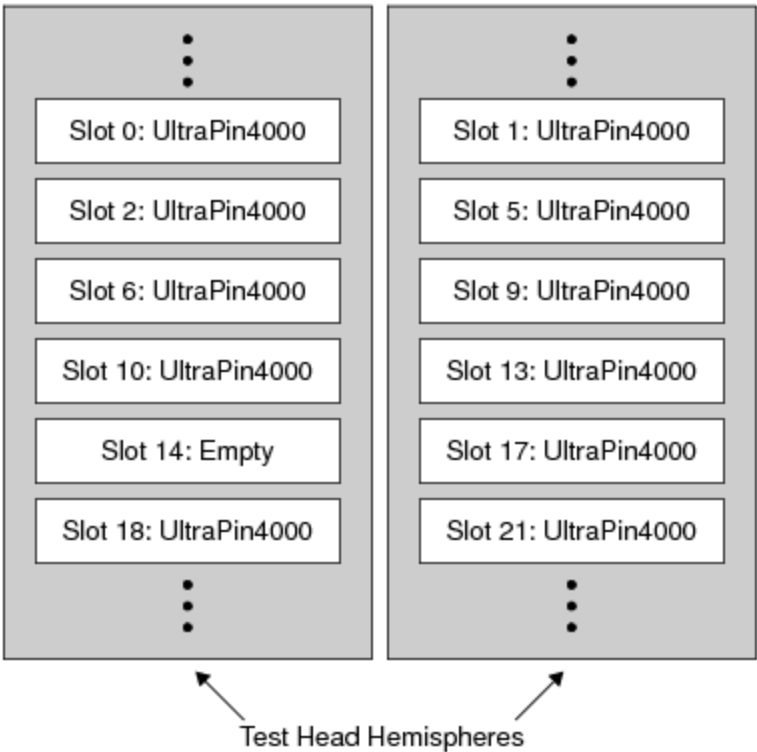
UltraPin4000 Slot-to-Slot Calibration Process

The UltraPin4000 supports an optional slot-to-slot calibration that provides improved timing accuracy and requires a special calibration DIB. This calibration provides a tighter EPA of 35ps between boards in up to four adjacent slots on one test head hemisphere. You must have a special calibration DIB to run this calibration (Teradyne part number 601-641-00 for 12 and 24-slot systems and 601-642-00 for 36-slot systems). Boards that are more than four slots apart or are on opposite test head hemispheres default to the system EPA of 125ps. If you have an empty slot or slot filled with any board other than an UltraPin4000 between UltraPin4000 boards, then you cannot achieve the improved slot-to-slot EPA.

The figure below shows a slot configuration example where the EPA between the UltraPin4000 instruments is 125ps if slot-to-slot calibration is not run on the test system. If you run slot-to-slot

calibration, you get the following EPA:

- EPA between slots 0, 2, 6, and 10 is 35ps.
- EPA between slots 1, 5, 9, and 13 is 35ps.
- EPA between slots 5, 9, 13, and 17 is 35ps.
- EPA between slots 1 and 17 is 125ps because they are more than four slots apart.
- EPA between slots 10 and 18 is 125ps because slot 14 does not contain an UltraPin4000.
- EPA between slots 18 and 21 is 125ps because they are on different hemispheres.



UltraPin4000 EPA on Sample Slot Configuration

To enable this option follow these steps:

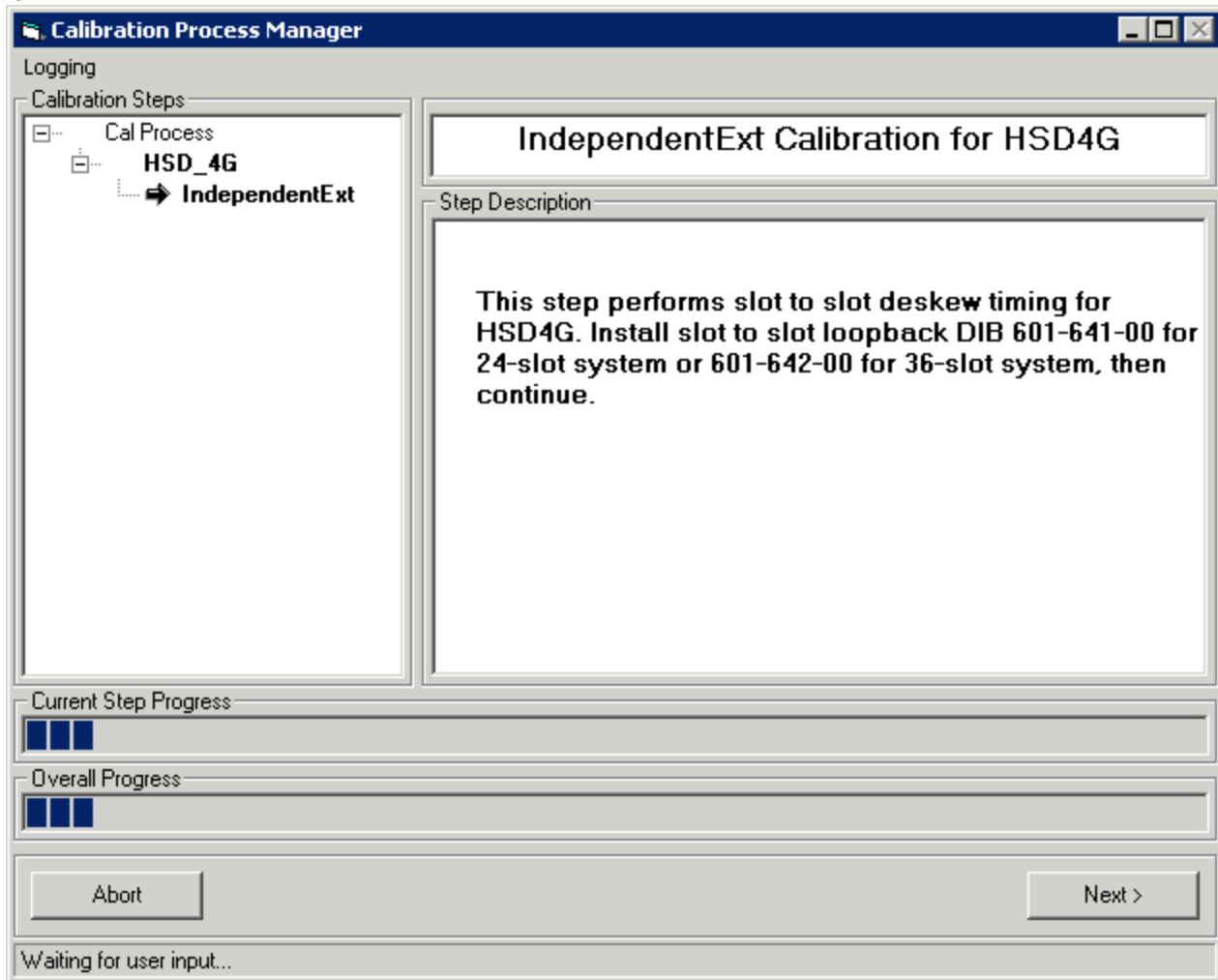
- Install a special calibration DIB.

Teradyne part numbers for these DIBs are 601-641-00 for 12 and 24-slot systems and 601-642-00 for 36-slot systems.

2. Once the DIB is in place, run independent external calibration in the Calibration tab of the IG-XL MUI.

If an incorrect DIB is installed or no is DIB present, then the Calibration Process Manager reminds you to install a valid calibration DIB, as shown in the figure below. Click Next to proceed.

If you click Next before installing a valid calibration DIB, then the DIB Down Detection calibration section fails, and the calibration run status reports EE (external equipment failure), rather than P (pass).



Calibration Process Manager Message for HSD4G Slot-to-Slot Calibration

✓ **Note:**

You must recalibrate UltraPin4000 instrument boards whenever you remove or replace any individual UltraPin4000 board. This rule also applies to a board that has been removed and replaced in its original location, which is different from normal independent calibration.

Recalibration is required because slot-to-slot calibration regroups the configuration based on adjacent slots after any UltraPin4000 instrument has been removed or replaced.

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Digital Instrument Calibration : Running Job Dependent Calibration

Running Job Dependent Calibration

Job dependent calibration must be executed after a test program has been loaded and the DIB has been mounted on the test head. Dependent calibration is executed from IG-XL after the loaded test job has been validated.

 To run, select User DIB Calibration from the Teradyne IG-XL menu or click Calibrate TDR in the Run group on the IG-XL Main tab.

 Note:

Make sure no devices are in DIB sockets before running User DIB Calibration.
User DIB calibration uses time domain reflectometry (TDR) to determine the trace lengths of all signals used in the test program. This calibration runs without any status messages and takes one minute or less for typical test programs. If any channels fail for any reason, those channels are listed in the test program output window.

Removing Pins from Job Dependent Calibration

By default, IG-XL calibrates all digital pins in the test program channel map. You may find it useful, however, to remove certain channels from the list of channels used by a job during calibration. Normally, you do this when the DIB has some active circuitry and job dependent calibration fails. You can also do this when debugging calibration or job problems.
For instance, some DIB traces may have components attached to them or be tied to other traces on the DIB. These pins usually fail job dependent calibration since TDR generally does not work in these situations. In these cases, do not run job dependent calibration on these pins so that the overall calibration can pass.

To exclude a list of channels from job dependent calibration, use one of the following methods:

- Channel-based VBT property, TheHdw.Digital.Calibration.Channels.Excluded

Note that this VBT function is the preferred method for excluding channels.

You must call this property in an OnProgramLoaded exec interpose function. Include the list of channels you want to exclude in the channel list.

```
Sub OnProgramLoaded()
```

```
    TheHdw.Digital.Calibration.Channels("0.ch26").Excluded = True
```

```
End Sub
```

Note that each call to this statement replaces any earlier calls to this statement. So, to exclude multiple pins, make sure to group pins into a single call.

To include channels that were previously excluded back into job dependent calibration, set the flag to False, and if the test program has already been validated, then you must also call the TheHdw.Digital.Calibration.Validate method.

For more information, see [TheHdw.Digital.Calibration.Channels.Excluded](#) in the VBT Syntax Reference.

- [Pins-based VBT property](#), TheHdw.Digital.Pins.Calibration.Excluded

You must call this property in an OnProgramValidated exec interpose function because IG-XL only maps pin names after validation. Include the list of pins you want to exclude in the pin list.

```
Sub OnProgramValidated()
```

```
    TheHdw.Digital.Pins("A0").Calibration.Excluded = True
```

```
End Sub
```

For more information, see [TheHdw.Digital.Pins.Calibration.Excluded](#) in the VBT Syntax Reference.

Once you execute one of these VBT functions, IG-XL removes the specified channels from the list of channels to be calibrated every time a program is loaded or validated, depending on the method in use. In addition, if the DIB EEPROM contains trace length data, calibration will not read the trace lengths for the channels listed in the function. If you need to change the list of channels to be removed, you must execute the VBT function again by either reloading the program or by running it manually from the Visual Basic editor.

Note:

In V7.30.00_uflx, the channel-based VBT syntax replaces the raw component object module (COM) calibration API. Teradyne no longer recommends using this API to exclude channels from job dependent calibration.

Trace Length Data in DIB EEPROM

All standard UltraFLEX DIBs contain a nonvolatile EEPROM circuit. Digital board job dependent calibration stores trace length data in the EEPROM by default after every passing calibration run. Once the data is stored, job dependent calibration does not need to be rerun on that DIB, unless you include additional channels that have not already been calibrated.

UltraFLEX DIBs are typically designed with four EEPROMs. A DIB should have at least one EEPROM, and the first EEPROM (EEPROM #0) is reserved for TDR calibration data, part numbers, and serial numbers. You cannot store any other information on this EEPROM. You can use additional EEPROMs (EEPROMs #1-3) for storing other data types. When you have EEPROM #0, you only have to run TDR calibration once, and IG-XL will load the TDR data from EEPROM #0 on subsequent job loads.

If you do not have EEPROM #0 on the DIB, then you must either run TDR calibration after every job load, or save the TDR file after the first TDR calibration run and reload that TDR file after every job load. You can only use the second option of reloading the TDR file if EEPROM #0 does not exist; otherwise, IG-XL loads the calibration data from EEPROM #0 and ignores the TDR file data. The example below shows how to set a TDR calibration data file path when you do not have EEPROM #0. IG-XL loads the actual TDR calibration data in this file during job validation.

Sub OnProgramLoaded()

```
TheHdw.Digital.Calibration.Instrument(tlCallInstrtumentHSD1000) _
    .DIB.DataFile = "C:\temp\hsdm_data_file"
```

End Sub

Copy the TDR calibration data file stored in the temp directory to your test program directory because this file is overwritten during every TDR calibration run. For information on enum values used with this property, see TheHdw.Digital.Calibration.Instrument.DIB.DataFile.

When TDR calibration fails on a specific channel, IG-XL writes a value of 65534ns to EEPROM #0 for that channel. IG-XL does not treat this value as a trace length but as a flag that the channel has failed calibration. If a channel is never calibrated, IG-XL writes a value of 65535ns to the EEPROM. Note that you may get unreliable results for any channel that was not calibrated or that failed TDR calibration.

When using a DIB that contains trace length data in its EEPROM circuit, the data is automatically loaded at the end of job validation. This process takes several seconds. Devices can be in the sockets during this time, since calibration is not being run.

A passing or failing user DIB calibration run overwrites any existing data that is in the EEPROM with the new calibration data.

You can also read and write trace length data in the DIB EEPROM. This is useful when some channels cannot be calibrated because of the DIB design, but trace length offset is desired for timing accuracy.

The following example shows how to write the trace length values.

Sub WriteTraceLength()

```
' Store same trace value for all excluded channels
TheHdw.Digital.Calibration.Channels("1.ch4 3.ch5").DIB.Trace = 9e-9
```



```
' Store different values for different channels
TheHdw.Digital.Calibration.Channels("1.ch4").DIB.Trace = 9e-9
TheHdw.Digital.Calibration.Channels("3.ch5").DIB.Trace = 12e-9
```

End Sub

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Digital Instrument Calibration : Calibration Dependencies

Calibration Dependencies

For a test program to run in production mode, the test system must have valid values for all calibration types. Independent calibration must be passing for all channels in the job for job dependent calibration to pass. All channels in the job must also pass job dependent calibration. The status of calibration will be failing if any of the following conditions are true:

- Job dependent calibration fails on one or more channels in the current job
- Independent calibration fails for any channels used in the current job
- Independent calibration has not been run for one or more channels in the current job
- Any combination of the above conditions

Channels failing independent calibration are ignored if those channels are not used in the current test program.

Any failures in independent calibration appear as failures during job dependent calibration, even if the job dependent calibration portion has passed. Failures are reported in the test program output window at the end of job dependent calibration. For each channel, a number of single-letter codes indicate its status:

P	Channel passed all calibrations.
F	Channel failed job dependent calibration.

C	Channel passed all calibrations, but values were close to failing limits on one or more tests.
D	Channel has broken dependency (for example, failing or missing independent calibration data).
U	Channel was untested.

If any channels fail the independent calibration portion of a job dependent calibrating run, a popup window appears and indicates which boards have failed. You then have the option to abort the job run or to proceed using nominal values.
For more information, see Nominal Values.

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Digital Instrument Calibration : Nominal Values

Nominal Values

Nominal values are values used by IG-XL calibration when passing calibration values are not available. This condition can occur because a calibration failed or was not run or its values have become invalid. Calibration attempts to use the best data available when loading nominal values. In general, all calibration results are loaded and used, even if the result is a fail. The reason for this behavior is that failing calibrations usually fail specific tests on specific channels. This means that you may have calibration values that will work for certain tests and channels.

For example, you may have a situation where job dependent calibration passes, but independent calibration fails some timing calibrations on a handful of channels. In this situation, IG-XL loads both the passing job dependent calibration data and all independent calibration data values (both passing and failing). With this setup, you may be able to run PMU tests without any problems, or you may be able to disable a site to avoid channels with independent channel failures.

When no data exists for any calibration, the use of nominal values means that default values are loaded. Under these conditions, test results will most likely be invalid, especially if no independent calibration data exists. For example, running without job dependent calibration data means that trace length values are assumed to be zero, and any functional test will probably fail.

IG-XL notifies you at run time whenever nominal values are used. Since the use of nominal values indicates that not all calibrations are passing and available, you can decide if you want to continue or if you need to calibrate the test system.

	
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Digital Instrument Calibration : Calibration Data Storage and Status Displays

Calibration Data Storage and Status Displays

Calibration Data Locations

Calibration data for each of these data types is stored in a different location.

- Independent calibration data is stored in files on the host computer.
- Job dependent calibration data is stored on the DIB EEPROM circuit.

Independent calibration data files for all calibration runs are stored on the host computer. IG-XL generates a data file per board that is named using calibration type, date/time stamp, instrument name, slot number, and a date/time stamp in this format:

<Cal ID>_<Date and Time>_<Instrument Name>_<Slot Number>.<extension>
where the extension can be .aelsm or .calarc.

The LatestCalibration folder also contains data for the last calibration run. Files are stored in the following path:

C:\Program Files (x86)\Teradyne\IG-XL\Cal\HSD-M | HSD-I | HSD-4G | HSD-U

In this folder, IG-XL creates a new time-stamped folder for each calibration run for the last 10 runs. Folders are named by date and time of execution. In addition to data files, IG-XL stores status files and calibration output and configuration files here.

Job dependent calibration data is stored in the DIB EEPROM as discussed in Running Job Dependent Calibration. Every time a test program is validated, job dependent calibration data must be reloaded from the EEPROM. If no data exists, job dependent calibration must be executed again, or the job can be run with nominal values. DIBs with passing calibration values in the EEPROM can be used on other test systems without requiring a new calibration.

Calibration Status Display

Since independent calibration is usually run during scheduled maintenance and not during normal testing, you may not be aware of the calibration status of the boards in your test system. IG-XL provides a calibration status window that tells you what calibrations are currently being used by the test program and whether those calibrations are passing.

To open the calibration status display, click **Tester Calibration Status** in the **Calibration Tools** group on the **IG-XL Tools** tab.

Calibration Status					
Board/Cal Type	Status	Cal Temp	Cal Date	Next Cal Date	
+ ○ DC30Board					
+ ○ HDVS1					
+ ○ HexWSBoard					
- ○ HSD4G					
- ○ Slot 0					
✓ FactoryInt	OK	N/A	7/10/2009 11:07:07 AM	Never Expires	
✓ IndependentInt	OK	N/A	1/8/2010 1:45:09 PM	4/11/2010 1:45:09 PM	
✓ FactoryCalTool	OK	N/A	7/10/2009 12:02:52 PM	Never Expires	
✓ FactoryLoopback	OK	N/A	7/10/2009 12:48:05 PM	Never Expires	
✓ IndependentExt	OK	N/A	1/8/2010 1:49:03 PM	4/11/2010 1:49:03 PM	
○ Dependent	Not Loaded	N/A	Not Available	Not Available	
+ ○ Slot 1					
+ ○ Slot 2					
+ ○ Slot 5					
- ○ HSDMBoard					
- ○ Slot 6					
✓ Independent	OK	25.2	11/5/2009 2:06:34 PM	2/6/2010 2:06:34 PM	
✓ Clock	OK	N/A	7/9/2009 9:57:36 PM	Never Expires	
○ JobDep	Not Loaded	N/A	Not Available	Not Available	
+ ○ Slot 7					
+ ○ Slot 8					

Calibration Status Display in IG-XL

The window displays all loaded calibrations for the current test program. It includes the slot, board type, calibration type, and when the calibration was run. If you open the window before a job is validated, job dependent calibration does not appear in the window.

If trace length data exists on a DIB that has been mounted on the test head, dependent calibration data appears after validation. If no data exists in the DIB EEPROM, dependent calibration data appears after user DIB calibration has been run.

Right-clicking a line in the window allows you to see details for the calibration. This feature is useful for independent calibration, as the details show the pass or fail status of all channels on that board.

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Manual Calibration

Manual Calibration

From time to time you may want to perform calibration at a specific point. In some cases, this is required. For example, you must perform user DIB calibration each time you change the device interface board (DIB). You can use the Calibration tab in the Maintenance Environment to start calibration manually. (You perform user DIB calibration after you load the job.)

Some test system elements use external components during their calibration procedures. You must calibrate these test system elements manually because managing these components interrupts test program flow and requires that you interact with the test system.

Use manual calibration when you want to:

- Calibrate test system elements that require external standards, such as a user or calibration DIB
- Perform calibration outside the autocal schedule

Manual calibration differs from autocal in the following ways:

- Manual calibration requires user interaction with the software or the test system and requires that you unload your test program. Autocal happens invisibly and does not require you to take any action.
- Manual calibration uses external reference standards (a calibration DIB or the DIB for your device). Autocal uses standards internal to the test system.

- [Calibration intervals are longer for manual calibration than for autocal.](#)
- [When IG-XL determines that manual calibration is necessary \(such as when the calibration interval for an instrument has expired\), IG-XL issues a warning but does not interrupt the job flow, allowing you to perform the calibration at your discretion. Autocal can be performed between job runs at the discretion of the software.](#)
- [Manual calibration can extend to the DUT socket \(for example, TDR calibration\). Autocal does not extend past the test system instrumentation.](#)

This section includes the following topics:

- [When To Perform Manual Calibration](#)
- [External Calibration](#)
- [Maintenance Environment's Calibration Tab](#)
- [User DIB Calibration](#)

For more information on calibration, see:

- [About Calibration](#)
- [Automatic Calibration](#)



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Manual Calibration : When To Perform Manual Calibration

When To Perform Manual Calibration
You must perform manual calibration when:

- | | |
|--|--|
| | <ul style="list-style-type: none">The calibration interval for a board or system component has expired (for a list of calibration intervals, see Calibration Types) |
| | <ul style="list-style-type: none">You make physical changes to the test head (for example, you move a board to a different slot, or you replace a board with one that has a different serial number, even if it is of the same type) |
| | <ul style="list-style-type: none">You change the DIB (this requires at least TDR calibration) |
| | <ul style="list-style-type: none">You replace a board that has been repaired by Teradyne |

You can run any test, including those normally managed by automatic calibration, at any time. For a general procedure for performing manual calibration, see External Calibration.

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Manual Calibration : External Calibration

External Calibration

You must perform external calibration periodically to maintain the internal standards used for autocal. This is usually necessary when IG-XL alerts you that certain calibration packets have expired for one or more boards. It requires you to install a calibration DIB on the test head and, therefore, interrupts the test program flow.

The general procedure for performing external calibration is:

1. Install the calibration DIB on the test head.

2. Launch the Maintenance Environment.

For a standard IG-XL installation, select Start>Programs>Teradyne IG-XL Vx.yy.zz>Tools>Maintenance, where x.yy.zz is the version of IG-XL you are currently running. You must exit IG-XL before you can launch the Maintenance Environment.

3. Click the Calibration tab.

4. In the calibration test selection pane, select the types of calibration you want to perform. For information on choosing which tests to run, see [Selecting Calibration Types](#).

5. Click the Calibrate button to begin calibration.

If IG-XL asks you to confirm that you want to run external calibration, click the OK button to proceed. To skip the external tests, click the Cancel button. Calibration proceeds for all selected internal tests.

Calibration is complete when Calibration successful appears in the Output Window. You can verify the results in the Calibration Status pane.

If an external calibration run fails, you must resolve the failure and successfully calibrate the element before you can continue running your test program. In engineering mode, IG-XL lets you use nominal (default) calibration factor values to permit test program development. Take care, however, when interpreting results generated using nominal calibration factors.

- | | |
|----|---|
| 6. | When calibration completes and you have verified the results, exit the Maintenance Environment. |
| 7. | Replace the DIB for your device on the test head. |
| 8. | Run your test program as normal. |

You must still perform user DIB calibration on your device’s DIB before you can run your test program.

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Manual Calibration : Maintenance Environment’s Calibration Tab

Maintenance Environment’s Calibration Tab

Use the Calibration tab in the Maintenance Environment to calibrate specific test system resources (including those normally managed by autocal), to calibrate test system resources outside the scope of your test program, and to perform external calibration using a calibration DIB or a device-specific DIB.

The following topics are available:

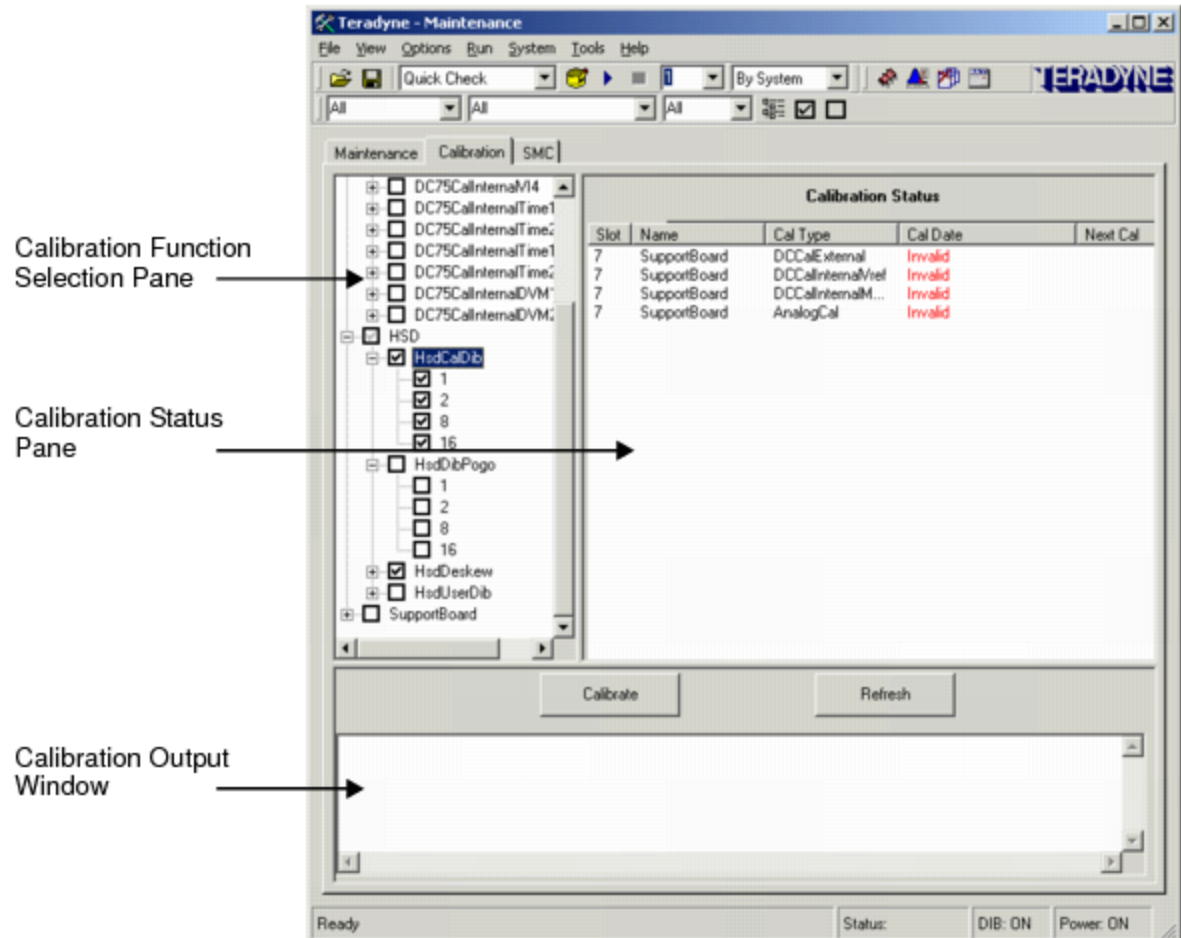
- Calibration Function Selection Pane
- Calibration Status Pane
- Calibration Output Window
- Selecting Calibration Types

Using the Calibration Tab

To use the Calibration tab in the Maintenance Environment:

- Select the test system resources you want to calibrate. (For more information on selecting test system resources, see Selecting Calibration Types.)
- Click the Calibrate button.

For information on performing a full external calibration using a calibration DIB or a device-specific DIB, see External Calibration.



Maintenance Environment Calibration Tab

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Manual Calibration : Maintenance Environment’s Calibration Tab : Calibration Function Selection Pane

Calibration Function Selection Pane

The Function Selection pane allows you to select the calibration functions you want to perform and the boards on which you want to run them. You can select tests by board or by test. Options in this pane are arranged in the following order:

- The top level lists all board types in the test system
- The second level, under each board type, lists calibration functions available for that board type
- The third level, under each function type, lists the test head slots containing that board type

For information on selecting calibration types, see [Selecting Calibration Types](#).
See also:

- [Selecting Calibration Types](#)
- [Calibration Output Window](#)
- [Calibration Status Pane](#)
- [About Calibration](#)

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Manual Calibration : [Maintenance Environment’s Calibration Tab](#) : Calibration Status Pane

Calibration Status Pane

This window shows results from the most recent calibration run. This includes the following information:

- | | |
|---|---|
| • | Board type, slot number, and calibration function |
|---|---|
- | | |
|---|---|
| • | Date of the most recent calibration |
|---|---|
- | | |
|---|--|
| • | When autocal expects to perform this function next |
|---|--|

The values that appear here are taken from the hardware, not from the calibration database. The text in this window is read-only.
See also:

- | | |
|---|---|
| • | Selecting Calibration Types |
|---|---|
- | | |
|---|---|
| • | Calibration Output Window |
|---|---|
- | | |
|---|---|
| • | Calibration Function Selection Pane |
|---|---|
- | | |
|---|-----------------------------------|
| • | About Calibration |
|---|-----------------------------------|

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Manual Calibration : Maintenance Environment’s Calibration Tab : Calibration Output Window

Calibration Output Window

This is a calibration-specific version of the Maintenance Environment’s Output Window. This window shows the current test status and any messages generated during the calibration run. To save the messages generated during a particular calibration run, copy the text from this window into a text editor and save the file.

See also:

- [Selecting Calibration Types](#)
- [Calibration Function Selection Pane](#)
- [Calibration Status Pane](#)
- [About Calibration](#)

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Manual Calibration : Maintenance Environment’s Calibration Tab : Selecting Calibration Types

Selecting Calibration Types

You can select calibration types in the following ways:

- Selecting by Board
- Selecting by Function

Selecting by Board

Normally, both autocal and manual calibration operate only on boards and channels that appear in a test program. To calibrate a board that a test program does not use, choose it in the tree. To force calibration for a specific board:

- On the Maintenance Environment’s Calibration tab, expand the function selection tree (the left pane) to show the calibration functions available for the board types you want to calibrate.
- For each function you want to perform, open the selection tree to show the list of applicable test head slots.
- Check the box to the left of each board you want to calibrate. Repeat this for each function you want to perform.
- Click the Calibrate button to begin calibration.

IG-XL performs calibration for the selected functions.

Calibration is complete when Calibration successful appears in the Output Window. You can verify the results in the Calibration Status pane.

See also:

- Maintenance Environment’s Calibration Tab
- External Calibration
- Manual Calibration

Selecting by Function

You can perform one type of calibration across specified boards of the same type, or across all boards of the same type.
To run all tests against multiple boards:

- On the Maintenance Environment’s Calibration tab, expand the test selection tree (the left pane) to show the board type you want to calibrate.
- Select the tests you want to perform.

Maintenance selects all boards under the test types you select. To exclude some boards from this test, open the tree for each test and deselect the boards you want to exclude.

- Click the Calibrate button to begin calibration. Maintenance performs the function for all selected boards.

Calibration is complete when Calibration successful appears in the Output Window. You can verify the results in the Calibration Status pane.

To perform the equivalent of an autocal, select all tests that do not include “DIB” or “External” in their names.
To perform a full external calibration, select all tests that include “DIB” or “External” in their names.
For a full reference of internal and external calibration functions, see Calibration Types.
For more information on calibration under IG-XL, see About Calibration.



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Manual Calibration : User DIB Calibration

User DIB Calibration

Before you run your test program, you must calibrate your DIB to allow IG-XL to adjust signal timing according to the characteristics of the DIB. This procedure measures the electrical path length from the end of the Pogo pins to the device contacts on the device interface board (DIB), handler interface board (HIB), or prober interface board (PIB). The calibration, based on the specific hardware interface in use on the tester, is used to apply a correction to the programmed received edges.

For digital channels, user DIB calibration is called TDR (time domain reflectometry) calibration. For each channel in your test program, TDR measures the path length of each channel in the time domain. That is, TDR measures the time it takes a signal to travel from the tip of the Pogo pin to the end of the contacts in the device socket on the DIB. This allows precise timing of edges based on the characteristics of the specific DIB.

Because user DIB calibration is dependent on the interface board, you must recalibrate each time you change the DIB or any time you load a job.

In production mode, user DIB calibration is required before executing a test program.

In engineering mode, user DIB calibration is optional. This allows you to use nominal calibration data.

IG-XL stores user DIB cal factors in the following folder:

\<installation-directory>Teradyne\IG-XL\Cal\<instrument-name>

When you restart DataTool or load a new job, IG-XL unloads the existing calibration factor files. IG-XL creates new calibration factors when user DIB calibration is performed. The values in this file represent the path delay (one-way) from the DIB Pogo pin to the DUT.

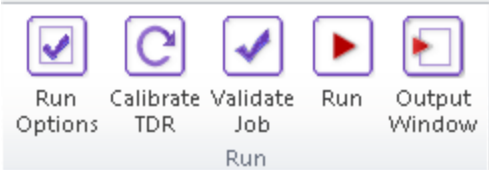
Running User DIB Calibration

Before performing user DIB calibration, you must verify that job-independent calibration on all channels in your test program have passed.

User DIB calibration must be performed with:

- | | |
|---|--|
| • | The test program loaded |
| • | The DIB/PIB engaged |
| • | All sockets open (no device in any socket) |

If the DIB/PIB is not mounted, or the sockets are not open, user DIB calibration fails.
To run user DIB calibration in engineering mode (with DataTool open), on the IG-XL Main tab, in the Run group, click Calibrate TDR:



DataTool Run Group

In production mode (when the program is loaded from an operator interface), the operator interface should include a button for user DIB calibration. See TDRCalButton Control in the Production Controls section.

Modifying User DIB Calibration

By default, calibration includes all pins. For information on modifying user DIB calibration, such as excluding a list of digital channels from job-dependent calibration, see Removing Pins from Job Dependent Calibration.

Instrument-Specific ConsiderationsBoards that calibrate out to the DIB often have special considerations regarding DIB circuitry, such as relays and capacitors. For information on instrument-specific user DIB calibration, click Related Topics.

